TEXORA · COURSE NOTES · LECTURE SHEET

Protein Fibres: Wool, Silk and Other Animal Fibres

The animal fibres built from protein — wool's scaly, crimped hair and silk's continuous filament, their structures, properties and care.

TE-1209 · NATURAL TEXTILE FIBRES · TOPIC 3 · LEVEL 1 TERM 2

1. The common chemistry: protein

Animal fibres are built from **proteins** — chains of amino acids joined by peptide bonds. Wool's protein is **keratin**, rich in sulphur-containing crosslinks (disulphide bonds) that give wool its elasticity and chemical character; silk's is **fibroin**, a simpler, more ordered protein that forms smooth filaments. Shared family behaviour follows from the chemistry:

- **Acids are tolerated, alkalis attack.** The mirror image of the cellulosics: protein fibres withstand dilute acids (acid dyes are used on both wool and silk) but are damaged — wool severely — by alkalis, which break peptide and disulphide bonds.
- **High absorbency.** Protein chains carry many water-attracting groups; wool's regain of roughly 13–16% is the highest of the common fibres, and silk's is about 11%.
- **Burning:** they burn slowly, self-extinguish when the flame is removed, and smell strongly of burning hair, leaving a crushable black bead.
- **Biology:** they are food for moth larvae and carpet beetles — the reason for mothproofing — and mildew attacks them less readily than cellulose.
- **Heat:** prolonged heat and strong sunlight degrade them; wool yellows and silk weakens (silk is notably sensitive to light).

2. Wool — the hair of the sheep

Wool is the fleece fibre of sheep, shorn once or twice a year, then **scoured** to remove grease (lanolin), suint and dirt before processing. No other common fibre has its combination of warmth, elasticity and resilience, and the reasons all lie in its structure.

2.1 Structure

A wool fibre is a biological composite built in three regions. The outer **cuticle** is a layer of overlapping **scales**, like roof tiles, pointing toward the fibre tip — visible under the microscope and unique to animal hair fibres. Beneath it, the **cortex** forms the fibre's body: long spindle-shaped cells whose two halves differ slightly in chemistry, and this bilateral structure is what makes the fibre curl into its natural **crimp**. Coarse wools may add a central hollow core, the **medulla**. Crosslinked keratin inside the cortex lets the fibre stretch by up to roughly a third of its length and spring back — the molecular basis of wool's wrinkle recovery and durability.

2.2 Properties and the felting phenomenon

- **Warmth:** crimp traps still air in bulky yarns and fabrics, and wool even releases a little heat as it absorbs moisture — the warmest common apparel fibre.
- **Elasticity and resilience:** the best of the natural fibres; wool garments shed wrinkles and carpets recover from foot traffic.
- **Absorbency:** regain 13–16%; wool can take up a large quantity of water before feeling wet.
- **Strength:** modest, and lower when wet; abrasion resistance is good because the fibre bends rather than breaks.
- **Felting:** wool's one notorious weakness. Under heat, moisture, alkali and agitation, the surface scales make fibres migrate root-ward and lock together irreversibly — the fabric shrinks and mats into felt. The same property is used deliberately to make felt and to finish woollen cloth (milling), and fought with anti-shrink treatments in machine-washable wool.

> **WORKED REASONING — WHY WOOL FELTS AND SILK DOES NOT**
>
> Felting needs two ingredients: a scaly surface that lets a fibre slide one way more easily than the other (the directional friction effect), and conditions — warmth, moisture, agitation — that make fibres move. Wool has both scales and crimp, so in a hot wash its fibres creep, entangle and lock. Silk's filament is smooth, scaleless and nearly straight: there is no ratchet to drive migration, so silk fabrics do not felt (they have other weaknesses instead — to light, perspiration and abrasion). The microscope, in this case, predicts the laundry.

Uses: suiting and knitwear, blankets, carpets (resilience), upholstery, felt, and technical felts and insulation.

3. Silk — the continuous filament

Silk is the fibre of the cocoon of the silkworm (*Bombyx mori*), raised on mulberry leaves in the ancient industry of **sericulture**. The caterpillar extrudes two fibroin filaments cemented together by a gum, **sericin**, forming a cocoon of a single thread that may run to a kilometre or more. In the filature,

cocoons are softened in hot water, the thread ends found, and several filaments reeled together into raw silk yarn. Boiling off the sericin (**degumming**) in finishing releases silk's famous lustre and softness — and removes roughly a fifth to a quarter of the weight.

3.1 Properties

- **Form:** a true filament — smooth, triangular in cross-section, which refracts light and gives silk its natural lustre.
- **Strength:** the strongest natural fibre for its fineness, with good elasticity (though less recovery than wool).
- **Handle and drape:** smooth, cool, fluid — the luxury standard; also an excellent insulator for its weight.
- **Weaknesses:** degraded by sunlight and perspiration, damaged by alkali, expensive, and produced by a labour-intensive process; wild silks (tussah, eri, and the golden muga) are coarser, less lustrous and hardier than the cultivated mulberry silk.

Uses: luxury apparel and sarees, ties and scarves, lingerie, high-value furnishings, and surgical sutures (a reminder of silk's biological compatibility).

4. The other animal fibres

FIBRE	SOURCE	CHARACTER AND USE
Cashmere	Downy undercoat of the Kashmir goat	Extremely fine, soft and warm; combed out in small quantities — hence its price; luxury knitwear.
Mohair	Angora goat	Lustrous, resilient, longer hair fibre; suiting and upholstery blends.
Alpaca	Alpaca (South America)	Warm, silky, low in grease; knitwear and coatings.
Camel hair	Bactrian camel undercoat	Fine, warm, naturally tan; classic overcoating.
Angora	Angora rabbit	Very fine, fluffy and warm but weak; used in small blends for softness.

These "speciality hair fibres" share wool's keratin chemistry but are finer, rarer and dearer; in trade they are usually blended with wool or synthetics and declared on labels as a selling point.

5. Wool and silk compared

FEATURE	WOOL	SILK
Form	Staple fibre, crimped	Continuous filament, straight
Surface	Overlapping scales	Smooth
Protein	Keratin (sulphur crosslinks)	Fibroin (+ sericin gum)
Signature strength	Elasticity, warmth, resilience	Lustre, fineness, drape
Signature weakness	Felting shrinkage; moth attack	Sunlight and perspiration damage; price
Moisture regain	~13–16%	~11%

6. Measuring quality: wool fineness and silk size

Wool is traded on fineness. The single most valuable measurement of a wool lot is its mean fibre diameter, in micrometres (microns): fine Merino wools run about 17–24 microns and go to apparel; medium wools of 25–32 microns to knitwear and blankets; coarse wools above 33 microns to carpets. (The older “quality number” or Bradford count expressed the same thing inversely — the finer the wool, the higher the count of yarn it could spin.) Two lots of equal weight can differ several-fold in price on diameter alone, because diameter decides softness, the finest yarn spinnable, and the fabric’s resistance to the prickle felt from coarse fibres. The second commercial number is **yield**: raw (greasy) wool loses grease, suint and dirt in scouring — often a third or more of its weight — so wool is bought on its expected clean yield, and staple length and vegetable-matter content complete the specification.

Silk is measured by denier. The size of a silk filament or yarn is expressed in denier — the weight in grams of 9,000 metres. A single cocoon filament is only about 2–3 denier, which is why the filature reels several together: the common raw-silk size of 20/22 denier is built from roughly seven to ten filaments, and any break or thin place shows immediately in so fine a yarn. Evenness of denier along the thread (its “size deviation”) is a principal silk quality grading factor, alongside cleanness and neatness of the reeled skein.

7. Glossary — key terms of this topic

TERM	MEANING
Keratin	The sulphur-crosslinked protein of wool and all hair fibres.
Crimp	The natural waviness of a wool fibre, arising from its two-sided cortex; it creates bulk and warmth.

Felting	The irreversible matting of wool under heat, moisture and agitation, driven by its surface scales.
Sericulture	The rearing of silkworms and production of cocoons for silk.
Sericin	The gum cementing the two fibroin filaments of silk; removed by degumming.
Scouring (wool)	Washing raw fleece to remove grease, suint and dirt before processing.

8. Key points and summary

- Protein fibres invert the cellulosic rules: tolerant of acid, damaged by alkali, highly absorbent, and smelling of burning hair in the flame test.
- Wool’s scales, bilateral cortex and crimp explain its warmth, elasticity — and its felting. Silk’s smooth triangular filament explains its lustre, drape and freedom from felting.
- The speciality hair fibres (cashmere, mohair, alpaca, camel, angora) trade in fineness and rarity, usually blended.
- Care follows chemistry: gentle, near-neutral washing for both fibres, protection from moths for wool and from sunlight for silk.

9. Practice

Answer in your own words; brief model answers follow.

1. Protein fibres are damaged by alkali but tolerate acid. Connect this to two practical facts: the dye classes used on wool, and the washing instructions on a wool sweater.
2. Draw (in words) the three structural regions of a wool fibre and state one property each region is responsible for.
3. A wool lot has a mean fibre diameter of 19 microns; another is 35 microns. Assign each a suitable end use and justify it.
4. Greasy wool loses “a third or more” of its weight in scouring. If a mill buys 12,000 kg of greasy wool at a clean yield of 62%, how much clean wool does it get, and why is wool priced on yield rather than on greasy weight?
5. Silk loses roughly a fifth to a quarter of its weight at degumming. A lot of raw silk weighs 80 kg before degumming; estimate the degummed weight range.
6. Explain why a raw-silk yarn of 20/22 denier is made from several cocoon filaments rather than one, using the denier of a single filament.
7. A customer complains that her wool cardigan shrank and stiffened in a hot wash but her silk blouse, washed the same day, did not felt. Explain both outcomes.

8. Why are the speciality hair fibres almost always sold in blends rather than as 100% products?

Answer key

1. Acid dyes are used on wool and silk because the fibres tolerate acidic dyebaths; alkaline washing liquor attacks peptide/disulphide bonds, so wool is washed near-neutral, cool and gently.
2. Cuticle (scales — friction, felting), cortex (keratin cells — strength, elasticity, crimp via its two halves), medulla in coarse fibres (stiffness, dyeing differences).
3. 19 µm — fine apparel knitwear/suiting (soft, spinnable fine, low prickle). 35 µm — carpets/blankets (resilience and bulk matter, fineness does not).
4. 12,000 × 0.62 = 7,440 kg. Grease, suint and dirt content vary between lots, so greasy weight is not comparable — clean yield is the real quantity bought.
5. About 60–64 kg (80 × 0.75 to 80 × 0.80).
6. One filament is only about 2–3 denier — far too fine and weak alone; reeling roughly 7–10 together builds a usable 20/22 denier thread.
7. Wool felts: heat, moisture, alkali and agitation drive its scaled fibres to migrate and lock. Silk filaments are smooth and scaleless — no ratchet, no felting.
8. Scarcity and price; blending keeps the signature softness/warmth while wool or synthetics supply bulk, strength and affordability.
